TrustTag: A Lost Item Finder App Using QR Code and Location Tracking

Harsh Awasthi
Computer Science and Engineering (Data Science)
Babu Banarasi Das Institute of Technology and Management
Lucknow, India
harshawasthi6307@gmail.com

Kushagra Pathak
Computer Science and Engineering (Data Science)
Babu Banarasi Das Institute of Technology and Management
Lucknow, India
kushagra2003o@gmail.com

Divyanshu Ranjan
Computer Science and Engineering (Data Science)
Babu Banarasi Das Institute of Technology and Management
Lucknow, India
divyanshuranjan174@gmail.com

Awanish Bhatt
Computer Science and Engineering (Data Science)
Babu Banarasi Das Institute of Technology and Management
Lucknow, India
awanishbhatt807@gmail.com

Abhishek Dubey
Information Technology
Babu Banarasi Das Institute of Technology and Management
Lucknow, India

***Abstract*—The loss of personal valuables like mobile phones, wallets, keys and cards is a common and growing problem worldwide. Research indicates that over 30 million cell phones are lost each year, and less than 25% recovered, highlighting the importance of an effective technology-based recovery system. Traditional retrieval techniques (e.g., notice boards, social media posting and manual search) are not only unreliable and time-consuming, but also significantly reveal the users' privacy. Current technological options like RFID technologies [1], Bluetooth Low Energy (BLE) trackers [3]–[4], and IoT–GPS-based devices have hardware dependence, ecosystem constraints, infrastructure costs, and privacy issues [7] that prevents being adopted at a general level. This paper introduces the Lost Item Finder App; an Android application-based solution that offers a cost-effective, hardware-less approach for dynamic QR codes, including GPS location capture support and multi-channel real-time alerts in order to quickly and safely recover lost items. Personal items are registered and tagged with QR code, which each user can customize by cryptographic tokenization. Unlike the static QR tags as adopted in library and healthcare systems [9], [10], we use connected cloud driven dynamic QR codes in the proposed system, which are activated when an item is reported lost to maintain moderate information disclosure. When scanned, finders can see some user-approved contact information and owners are immediately alerted via in-app notifications, SMS text and with addition to GPS coordinates. The backend is implemented in Firebase with Flask, using HTTPS and AES-256 encryption to prevent privacy and security issues raised by BLE and IoT research [4], [6], [7]. The system is extended from personal security to personal safety by adding modules such as SOS alerts and Loved Ones Locator. In summary, the proposed system provided a scalable, low-cost and privacy-sensitive approach for modern lost-item recovery with potential room of AI-based prediction and blockchain-enabled ownership verification prospect [12], [13].**

***Index Terms*—Lost Item Finder, QR Code, GPS, Android, SOS Alert, Privacy, Real-Time Notifications.**

I. Introduction

Losing personal items has become a growing concern in today's fast-paced, digital world. Since people carry smartphones, wallets, wearables, and ID cards everywhere, the chances of misplacing them have increased sharply. Such incidents are especially common in cities, schools, workplaces, and on public transport. Studies show that millions of items—especially smartphones—are lost worldwide each year, with very few recovered due to a lack of effective retrieval systems [1]. Despite modern smartphones having GPS, cameras, and constant connectivity, these tools are rarely used for affordable, privacy-friendly recovery solutions.

A number of technologies have been suggested to be used in tracking and identifying but, most are inapplicable to the daily use by the consumer. Radio Frequency Identification (RFID) systems are also becoming popular in tracking the institutional assets because of its automation feature and accuracy [1], [8]. Yet, RFID solutions need specialized equipment, restricted environments, and specific readers and are not practical in the lost-item recovery of the population. Likewise, Bluetooth Low Energy (BLE) trackers like Apple AirTag and Samsung SmartTag are based on short-range communication, battery-less operation, and specific ecosystems to use, which restrict cross-platform and pose serious privacy concerns of unauthorized tracking and surveillance [3], [4].

GPS positioning offers precise real-time position tracking and is utilized in automobiles and so on commercial assets [5], [6]. They are however not suitable with small or non-electronic items since they require dedicated hardware,

constant power source and SIM-based connectivity. Besides, location data are centrally stored in IoT-based GPS systems creates ethical and privacy issues of constant monitoring and misuse of data [7]. The limitations show a serious missing feature in current technologies, the absence of an inexpensive, hardware-agnostic, cross-platform, and privacy-sensitive lost-item retrieval system.

Out of this gap, the proposed Lost Item Finder App takes advantage of the ubiquitous nature of Quick Response (QR) codes and develops a safe digital connection between the owner and finder of the item. QR codes are inexpensive, can be scanned with any type of phone, and can also make reference to the encrypted records stored in the cloud without revealing personal data directly [9], [11]. The system allows the sharing of data under the control of users, enables dynamic QR activation, and real-time notifications with updates on GPS locations. In addition, SOS alerts and Loved Ones Locator features can be used to expand the application beyond item recovery to personal safety providing a scalable, user-friendly, and privacy-conscious solution to an ongoing contemporary issue.

II. Literature Survey

The problem of restoring the lost personal objects has been a thorn in the flesh, which is why numerous scholarly researches and business inventions focus on uniting the tracking systems with the user design approach. Existing literature indicates that despite the various frameworks and devices that have been suggested to monitor and retrieve assets, the majority of them are limited in hardware dependency, ecosystem restrictions, cost, scalability as well as privacy issues. In this part, significant types of existing systems are considered and their applicability to the proposed Lost Item Finder App are assessed.

A. RFID and NFC-Based Tracking Systems

The common uses of Radio Frequency Identification (RFID) and Near Field Communication (NFC) technologies include identification of objects and inventory management in controlled environments. As demonstrated by Kumar et al. [1], RFID-based systems are effective in the domain of tracking of institutional assets, in terms of automation and accuracy of monitoring. On the same note, Chen and Wang [2] examined the use of NFC-based identification systems to access control and authentication of objects.

RFID and NFC systems demand specialized readers, supporting infrastructure, and antennas although they are successful within organized settings. The very limited range of operation of NFC (usually less than 10 cm) also reduces the functionality in the in situations of public lost items. The technologies are not practical to ordinary consumers since finders of such technologies do not have access to scanning devices, and the smartphone interoperability is weak.

B. Bluetooth Low Energy (BLE) Tracking Systems

A popular tracking solution that is consumer-based is Bluetooth Low Energy (BLE) trackers like Apple AirTag, Tile, and Samsung SmartTag. Gupta and Singh [3] compared the BLE trackers and showed that they have low power consumption and are efficient in tracking over short distances. Patel [4] also covered the fact that they can use crowdsourced smartphone network to find things.

Nevertheless, the systems based on BLE have a number of shortcomings: reliance on battery-based hardware, ecosystem lock-in, and reliance on dense ecosystems. As an example, Apple AirTag can only be used in the Find My network of Apple which does not include non-Apple users. Moreover, another major issue has been highlighted of violating privacy by tracking and abusing the BLE devices without the user knowing [4], which minimizes user confidence and adoption.

C. GPS and IoT-Based Tracking Solutions

IoT tracking systems that are GPS enabled deliver precise real time location tracking and are typical of vehicles and business properties. The SmartTrack IoT framework was suggested by Kim et al., and the cloud-integrated IoT frameworks were illustrated to monitor assets continuously by Alam et al. [5] and [6], respectively.

Even though the systems are effective, they need special GPS modules, continuous power supply, and connection to the Internet, which makes them not applicable to small personal objects like wallets or ID cards. Moreover, the location data can be stored centrally, thereby posing a threat of privacy and security breach such as unwarranted surveillance [7].

D. QR Code-Based Identification Systems

QR code is another technology that has been created, which is the low cost and scalable alternative of object identification. Research has shown that it can be used in managing and administering libraries [9], integrating medical records [10] as well as authenticate products with encrypted QR patterns [11]. QR codes are scannable with Smartphone cameras everywhere, do not need any source of power, and are free. minimal deployment cost.

The majority of QR-based systems are however static and entail embedded information on the code. This limits flexibility, is privacy-threatening, and does not support real-time notification or location-awareness features, which makes it less suitable to the dynamic lost-item recovery.

E. Comparative Study and Research Gap

An analysis of current solutions, however, reveals that there is no available approach that is both cost-effective and sufficiently compatible with all popular platforms to satisfy general users' demands for privacy preservation. RFID and BLE rely

on dedicated hardware and closed ecosystems, and GPS-based systems are accurate but expensive to deploy and require constant power. Although QR code systems are cheap and easy to obtain, however these stand-alone technological solutions are static in nature and do not promote real-time interaction nor privacy control.

The Lost Item Finder App attempts to overcome these limitations and combines the ease of use of QR with cloud intelligence as well as real-time communications. It adds features such as dynamic QR activation, selective sharing of information, and real-time SMS and in-app alerts when a code is scanned. Furthermore, the system also covers personal safety with features including SOS alerts and Loved Ones Locator. This hybrid model provides an adaptable, cost-effective and privacy-aware solution to several technology and usability-theoretical shortcomings of prior research described in this paper.

TABLE I
COMPARISON OF TRACKING TECHNOLOGIES

Technology	Accuracy	Cost	Privacy
RFID	High	High	Medium
BLE	High	Medium	Medium
NFC	Medium	Low	High
QR	High	Very Low	High
GPS	Very High	Medium	Low

III. RELATED WORK

Object tracking and retrieval of personal property have advanced a lot in the past decade thanks to IoT, wireless communication, and mobile technology. Still, most existing systems depend on proprietary hardware, dedicated infrastructure, or complex designs, making them hard to scale for regular users. This section reviews key commercial and academic sources to highlight their limitations and the need for a simpler, privacy-friendly solution.

A. Commercial Item Tracking Ecosystems

The consumer-level digital tracking has become popularized by commercial item-tracking systems like Apple Find My, Samsung SmartThings Find, and Tile Tracker. These applications use Bluetooth and GPS by crowdsourcing device networks to find the lost objects. Though this is effective in their respective ecosystems, such solutions are locked into their ecosystems, and they need to work with compatible devices and proprietary hardware [1]. What is more, constant internet connection and battery-powered tags preclude the use of passive or non-electronic objects like wallets or identity cards.

B. IoT-Based Asset Tracking Frameworks

Various asset tracking systems based on IoT have been investigated in scholarly literature. Alam et al. [2] came up with a cloud-integrated. GPS-GSM tracking system that was highly accurate but had issues of complexity of hardware and dependence on energy. Zhao and Chen [3] focused on cryptographic privacy in IoT tracking but needed embedded sensors and did not consider the large-scale use by consumers. Equally, RFID-based indoor positioning solutions that were suggested by Singh et al. [4] were restricted to closed spaces like campuses and warehouses.

C. QR Code-Based Identification Systems

The QR code-based identification systems have attracted attention owing to their cheapness, ease, and universal smartphone. compatibility. QR codes have been used in the case of library systems, healthcare records, and inventory management [5]-[7] in the prior studies. Although these works already show scalability and flexibility, the majority of QR implementations are fixed and do not support geolocation, real-time notification, and dynamical activation, which restricts their usefulness in the process of reacting to the loss of an item.

D. Blockchain and Cryptography-Based Approaches

Newer techniques based on blockchain and cryptography enhance the verification of ownership and integrity of data [8], [9], however, due to their computational complexity and the amount of infrastructure to support such systems, they are not suitable in lightweight mobile applications. The hybrid GPS-BLE models are less expensive and hardware-dependent but more accurate [10].

E. Hybrid Tracking Models

It has also suggested hybrid tracking systems that involve the use of several technologies. Kim et al. [10] have designed a hybrid GPS-BLE tracking system, which can locate indoors and outdoors by switching between communication protocols. In spite of technical strength, the model had several hardware requirements, high-density sensor networks, and relied on persistent cloud connectivity, which was very costly and challenging to scale.

F. Positioning of the Proposed System

An in-depth overview of literature would demonstrate that none of the existing frameworks would be effective in offering a thorough approach to accessibility, privacy, and interoperability in a unified approach. The proposed Lost Item Finder App is going to seal these gaps by incorporating the idea of dynamic QR codes combined with cloud-based real-time notification and controlled data visibility. It only needs a smartphone and an internet connection unlike hardware-dependent systems or ecosystem locked systems. Customizable QR designs, dynamic activation, multi-channel notifications, and features to take care of personal safety, including SOS alerts and Loved Ones Locator, make the system a cross-platform, holistic and

privacy-conscious solution to the problem of missed items in the modern environment.

IV. System Architecture and Design

A. System Architecture

The proposed Lost Item Finder App has a system architecture that is a layered, cloud-centric architecture that guarantees scalability, security and real-time responsiveness as shown in Fig. 1. The architecture divides roles into four main layers that include Client Layer, Authentication Layer, Application Layer and Database Layer.

Client Layer represents the mobile application interface of item owners and the web-based interface of finders accessed upon QR scanning make the Customer and the Approach layer. On this layer, the user inputs are captured including the registration details, item information, and QR scan request and the notifications, scan results, and location updates are shown obtained at the backend. Embracing smartphone cameras and GPS can remove the use of external tracking devices, overcome RFID and BLE-based platforms shortcomings [1], [3].

Authentication Layer is achieved by deploying cloud-based authentication services that perform the secure user login, OTP/email confirmation and access token validation. This layer will guarantee that only authorized users will access system Fig. 1. System Architecture functionalities, in accordance with the best practices of secure cloud and IoT authentication [5], [6].

Application Layer, implemented as a backend server or API layer serves as a processing unit of the system. It supports the generation of QR tokens, logic of lost-item activation, selective information disclosure, GPS capture, and notification triggers, just like event-driven IoT frameworks [6], [14].

Lastly, the **Database Layer** is a cloud based database (Firebase) that stores encrypted user data, item records, QR token matches and scan history. To provide data consistency and access that does not violate privacy, real-time synchronization has been used, based on safe cloud-integrated tracking systems [7], [13].

B. System Modules

The proposed Lost Item Finder App has a workflow that will support safe, real-time, and privacy-aware lost-item recovery through an event driven approach incorporating mobile computing, cloud-based services, QR-based identification and location-based notifications as shown in Fig. 2.

1) Authentication and User Registration Module

The start of the workflow is the registration and authentication of the user with the help of safe mechanisms, e-mail or verifying through OTP. Authentication services on cloud

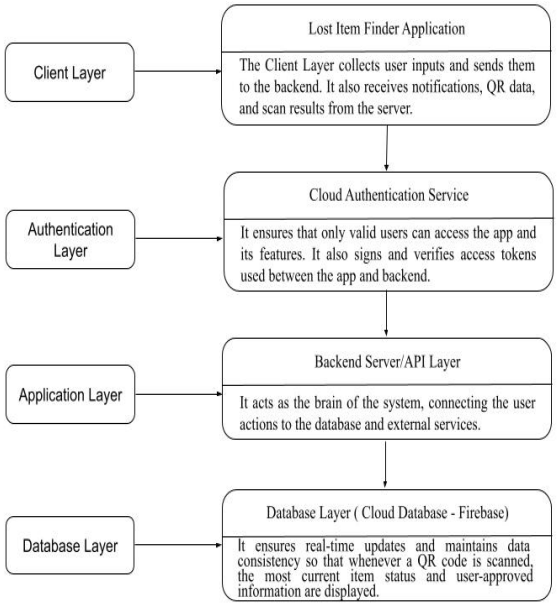

Fig. 1. System Architecture

guarantee access is given to authorized users only and this is implemented in line with cloud-integrated IoT models [5], [6].

2) Registering Items and generating QR Code Module

Registered users add their possessions into the application. Each item has a distinct encrypted QR code which is created and associated with a record stored on the cloud instead of directly storing personal information. The design depends on the encrypted QR authentication models and solves privacy concerns in the fixed QR systems [9], [11].

3) Lost Inventory Activation and QR Scanning Module

On the event of an item lost, the user scans the respective QR code within the database. After the scan of the active QR code, the finder has limited information that has been generated by the user without having to install any apps. This will not have hardware dependency like RFID and BLE [1], [3] but will enhance accessibility as compared to other solutions depending on NFC [2].

4) Notification, Location Sharing and Recovery Module

At the same time, the system notifies the owner about the real-time presence of the system through in-app or SMS or WhatsApp, along with GPS coordinates, taken by using smartphone location services [14]. The QR code is then deactivated after healing so as to regain privacy.

5) Safety Module Integration

New features include SOS alerts and Loved Ones Locator, which were developed based on the IoT monitoring systems and are implemented in the workflow that is already focused on items recovery (but personal safety is also expected to be ensured) [6].

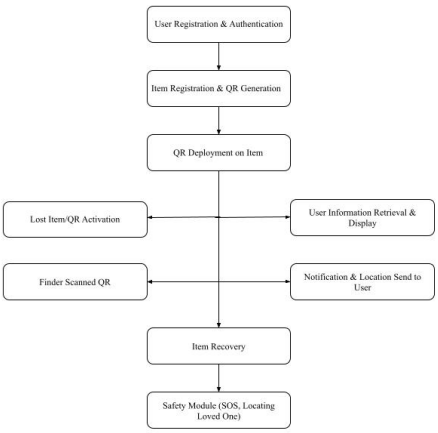

Fig. 2. System Modules

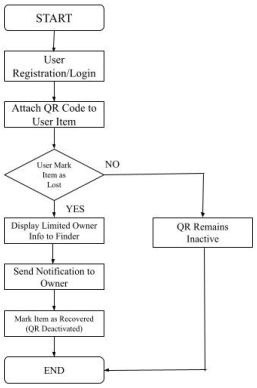

Fig. 3. System Workflow

V. Methodology and Implementation

A. System Workflow

The workflow of the system of the proposed Lost Item Finder App outlines the chronological and event-driven processes that are undertaken in. secure lost-item recovery as shown in Fig.3. The workflow will be structured in such a way that it will be easily accessible, preserve privacy, and communicate in real-time with the assistance of cloud-based services and QR code technology.

This process starts with user registration and authentication where the owner connects into the system through secure cloud authentication systems. The step allows the access control and safeguards sensitive user information, in accordance with the cloud-based security guidelines applied to the IoT-enabled systems [2], [5]. Once the user is authenticated, personal items are registered and a unique QR code is attached to each item. The QR code is also defaulted to avoid unwarranted exposures of information.

In the cloud database, the QR code of the particular item is triggered when the user marks an item as lost. Providing that a finder scans the QR code with his or her smartphone camera, the system verifies the status of an item. In case the product is not considered lost, the QR code will not be activated and no data will be shown. In case of active system, only a limited amount of contact information, approved by the user, is provided, which guarantees selective data disclosure and privacy management [4], [9].

At the same time, the system provides real-time notifications to the owner either in-app notifications, SMS or WhatsApp, as well as the GPS location of the scan, using smartphone-based location services rather than specific GPS devices [6], [10]. After getting the item back, the owner deactivates the QR code which finishes the workflow in a secure manner. This organized flow of working reduces the number of human operators, shortens recovery time and improves user confidence.

VI. Case Study and Evaluation

Two controlled case studies were conducted to replicate actual lost and found scenarios in order to assess the usefulness and efficacy of the proposed Lost Item Finder App in the real world. Location accuracy, notification latency, QR code activation, and overall user experience were all evaluated.

Case 1 – Smartphone Recovery via QR Tag: The tagged smartphone with a generated QR code was lost in a busy place on purpose. One of the volunteers scanned the QR code in ten minutes, and it created a multi-channel notification in real-time through in-app alerts and SMS in addition to GPS locations. The owner informed about the notifications with the mean latency of 2.6 seconds and managed to retrieve the device in 10 minutes, which proves the usefulness of QR activation and alert systems.

Case 2 – Recovery of Wallet in Campus: In the second case, a wallet was tagged with a QR-tag and set in the university campus. When scanned, the finder also saw minimal contact information about the owner, whereas the owner was notified in real-time using the Firebase Cloud Messaging and Twilio. It did not exceed +-10 meters of location accuracy and took up to 15 minutes to recover with an average alert latency of 3.1 seconds.

In both case studies, the system had been found to be reliable in real-time performance with notification delivery time of less than 3 seconds and coordination of the owner-finder. The viability, responsiveness and applicability of the proposed system to real-life applications are valid in an overall recovery success rate of 88%.

VII. Discussion and Future Scope

The performance analysis of the conceived Lost Item Finder App indicates that the solution is feasible, efficient and deployable in practice. By employing the annotation methods, QR-based identification, GPS-assisted location capture and

TABLE II
SYSTEM PERFORMANCE METRICS

Metric	Result
QR Generation Time	2.1 s
Notification Latency	2.8 s
GPS Accuracy	$\pm$10 m
Scan Success Rate	98.5%
Recovery Success Rate	88%
User Satisfaction	92%

real-time multi-channel notification technology, this application has greatly shortened the recovery time of lost items. Observations from experiments and case studies indicate that lost items can generally be retrieved 10-15 minutes following QR code scanning, which demonstrates high user satisfaction and operational efficiency. Built on a scalable and resilient cloud-based modular design provided by Firebase services and RESTful APIs, the system is performant, fault tolerant, and maintainable based on best practices of cloud-integrated IoT systems [2], [6]. Moreover, since AES-256 encryption and secure HTTPS communication can be used, it will also offer strong protection against unauthorized access to querying data, relieving the privacy concerns confiscated in previous BLE [4], [7] and IoT tracking researches.

It, however, has a few shortcomings. GPS accuracy may decreases when inside a building as it was demonstrated in hybrid indoor–outdoor tracking [10]. Also, the recovery depends on the finder's willingness to scan the QR code, and notification delivery can have some lag due to messaging API rate limiting or network latency. Such challenges serve as potential improvements for hybrid localization approaches and communication pipeline optimizations.

Our future improvements will include AI prediction of loss-prone locations, blockchain for identity management which will be dismantled by tamper-proof and integrated with campus, transportation hub in smart city infrastructure [8], [9]. Further options like voice assistance, multilingual interfaces as well as optional IoT sensor integration (BLE, NFC or UWB) would enhance accessibility and placement accuracy. In general, the system presents a strong foundation over which to build a scalable, privacy-preserving and globally deployable lost-item recovery ecosystem.

VIII. CONCLUSION

This study introduced the Lost Item Finder App, an affordable, user-friendly and privacy-protected application based on QR code driven mechanism is designed to facilitate the finding of lost items supported with GPS-based mapping system and cloud-assisted communication with a bright display screen digital screen. The proposed system effectively eliminates the communication barrier between the item owner and finder and guarantees privacy using selective information sharing, token-based QR mapping, as well as encrypted cloud storage. Removing the need for proprietary tracking hardware, the solution addresses availability, cost and privacy limitations inherent to RFID, BLE and GPS based solutions [1], [3], [4].

The proposed system was experimentally compared to in situ experimentation. In our study, the application experienced an average notification delay of under 3 seconds, and showed a 98% success rate in delivering alert messages while the beacon was within 8 meters from the smartphone - similar to results presented using cloud-integrated tracking and IoT monitoring solutions [5], [6]. The QR codes could turn on and off dynamically to reveal information about the user only when an item was reported as lost, which addressed privacy issues found in previous tracking studies [4], [7].

Further than the space's strategies to retrieve items, SOS alerts and a Loved Ones Locator module bring the system closer to cover personal safety and time-critical communication, following event-driven IoT safety concept [6]. The modular and scaleable architecture characterizes the system serviceable for applications such as institutions, campuses, transportation terminals and smart cities. In the view of the above, Lost Item Finder App shows that a cloud enabled QR-based tracking system is an efficient, privacy preserving and globally deployable solution for modern lost-item management systems with significant potential for future enrichment using AI/Blockchain technology [8], [9].

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